

KAREEM HATEM | SOFTWARE ENGINEER

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[LinkedIn](#) | [GitHub](#) | [Codeforces](#) | [ICPCID](#)

EDUCATION

Ain Shams University | Cairo, Egypt

October 2022 – August 2026

Bachelor of Computer and Information Sciences

- CGPA: 3.465/4.0

EXPERIENCE

Luciq (formerly Instabug) | Backend Developer Intern - Cairo, Egypt

July 2025 – September 2025

- Contributed to production code focused on cost and performance optimization.
- Reduced high-frequency API calls' latency by 90%, which improved the total average service runtime by 15%.
- Led the sunseting of a resource-heavy service and integrated its functionality into an existing service with no noticeable increase in CPU or memory usage, eliminating the cost of running the deprecated service.

PROJECTS

FOS - A Simple Operating System | C, Assembly

October 2024 – December 2024

- Implemented kernel and user heap memory allocation, increasing allocation speed by 85% and reducing memory usage by 25% through data structures.
- Developed efficient page fault replacement using the Nth clock algorithm, enhancing handling speed by +50% and reducing page faults by up to 10% (compared to FIFO).

C-DS - Data Structure Visualization | C++, ImGui

March 2024 – May 2024

- Developed interactive, real-time visualization modules for data structures and algorithms, enhancing user comprehension and improving newcomer learning efficiency by 40%.
- Incorporated OOP to ensure scalability and modularity, supporting future app and module expansions.

Image Encryption and Compression | C#, Windows Forms

March 2024 – May 2024

- Implemented image encryption and decryption utilizing Linear Feedback Shift Register (LFSR).
- Reduced file sizes by up to 60% while preserving image quality using Huffman Coding.

Dr Stone: Civilization Rebooted | C++, SFML

March 2023 – May 2023

- Led a team to deliver a high-quality project that won 1st place in the Freshman Best Projects Award.
- Integrated graph algorithms for path-finding and custom data structures, optimizing real-time game updates by 25%.

SKILLS SUMMARY

Programming Languages: C/C++, C#, Java, Python, Go, Ruby, JavaScript, TypeScript, Scala

Frameworks/Technologies: NodeJS, Rails, Linux

Tools: Git/GitHub, SQL, ClickHouse, HTML/CSS, Docker, Kubernetes, Bash

HONORS AND AWARDS

ICPC Regionals Finalist (2025 ICPC Arab and Africa Collegiate Programming Contest)

15th place and bronze medalist in the 2025 ICPC Egyptian Collegiate Programming Contest.

August 2025

1st place in the 2025 ICPC ECPC Qualifications Collegiate Programming Contest.

August 2025

18th place in the 2024 ICPC Egyptian Collegiate Programming Contest.

July 2024

1st place in the IBM Structured Programming Best Projects Award 2023 at Ain Shams University.

July 2023

EXTRACURRICULAR ACTIVITIES

acmASCIS Student Chapter

Training Committee Member

October 2023 – present

- Lectured and mentored 150+ trainees in algorithms and data structures for competitive programming.
- Designed, set, and tested over 40 problems used in acmASCIS training and local contests.

Training Committee Head

September 2024 – September 2025

- Lead a team of 25+ mentors to deliver competitive programming training to over 400 trainees.
- Coordinating problem-setting, testing, and resource teams to produce high-quality, open-source training materials.